

TABLE I
Different Types of Switches and Their Suitability for Different Application Needs

Control type	Suitability for tasks involving:			
	Speed	Accuracy	Force	Range
Push button	Good	Unsuitable	Unsuitable	Unsuitable
Toggle switch	Good	Unsuitable	Unsuitable	Unsuitable
Rotary selector	Good	Good	Unsuitable	Unsuitable
Knob	Unsuitable	Fair	Unsuitable	Fair
Small crank	Good	Poor	Unsuitable	Good
Large crank	Poor	Unsuitable	Good	Good
Wheel	Poor	Good	Fair/poor	Fair
Horizontal lever	Good	Poor	Poor	Poor
Vertical lever (to/from body)	Good	Fair	Short: Poor Long: Good	Poor
Vertical lever (across body)	Fair	Fair	Fair	Unsuitable
Joystick (lever)	Good	Good	Poor	Unsuitable
Pedal	Good	Poor	Good	Unsuitable



Fig. 2: Some discrete control elements

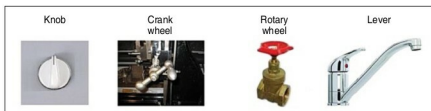


Fig. 3: Some continuous control elements

Effective controls have four important characteristics:

Accessibility. A control needs to be located in such a way that it can be reached easily and operated comfortably.

Identifiable. A control needs to be identified in terms of what it does in order to operate.

Functional. What a control controls and currently in what state it is needs to be indicated to the user when he or she looks at it. For example, a power-on toggle switch indicates whether it is on or off. This is useful especially when the product does not have a display.

Usability. A control should be easily usable with required force, speed and accuracy. This aspect is very critical, especially in continuous control, where a system's response is dependent on the control of the overall system.

Table I gives a concise view of the different types of switches and their suitability for different application needs.

With the availability of powerful processors and low-cost display technologies, a new input device available these days is the touchscreen, where keys are actually images created by graphics, and the overlay

touch mechanism, combined with the software, gives key-like input.

In addition to regular pushbutton switch based keys and touchscreens, another option that is available is known as membrane keys. These are made of PVC sheets of membrane with conductive elements sandwiched to form the key mechanism. The advantage of these keys is their form and shape that can be customised. Their low profile can be less than 1mm thick. Fig. 4 shows the types of membrane keys available and their construction.

While we have seen some critical design inputs, we also need to understand the basic characteristics of different switch types. There are four essential characteristics of switches:

Key size. Determines the ease with which fingers can be used.

Tactile feedback. Feedback that the user gets when the switch is closed, either by Click sound or haptic feedback to the finger pressing the switch.

View. Ability to view at night (whether keys can be illuminated or not).

Number of operations. A very important parameter that most designers miss out. Designers have to estimate the approximate number of times the keys will be used. Based on that estimate, the type of keys can be selected.

Mechanical keys tend to have large operating cycles, while membrane keys have the least. Many designs have failed due to the wrong usage of keys.

Table II gives the minimum distance that we should have between different keys as a matrix. This can help designers decide the spacing between different types of controls/switches by specifying various combinations of different switches.

Described below are the three types of keys:

Mechanical keys. These keys typically come in standard sizes and are excellent in terms of operational life and reliability. Depending on the working environment, the proper type